



Public Goods

Outlines

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Public Goods

- Public goods are goods or services that are non-excludable and non-rivalrous in consumption. Non-excludable means that individuals cannot be excluded from using the good, and non-rivalrous means that one person's use of the good does not diminish its availability to others. Classic examples of public goods include clean air, national defense, and street lighting.

Markets

- Market refers to any place that brings together buyers and sellers to exchange goods and services. These exchanges are typically driven by supply and demand, with prices determined by the interactions between buyers and sellers.
- Markets are crucial for resource allocation in an economy, as they efficiently distribute goods and services based on consumer preferences and producer capabilities.

Relationship between Markets and Public Goods

- **Market Provision of Public Goods**
- In many cases, public goods are not efficiently provided by markets because of their non-excludable nature. Since individuals cannot be excluded from consuming public goods once they are provided, there is a free-rider problem, where people may benefit from the good without paying for it. This leads to under-provision of public goods by the market.

Government Intervention

- Due to the market failure in providing public goods, governments often intervene to ensure their provision. This can involve direct provision by the government, subsidies to encourage private provision, or regulations to internalize the externalities associated with public goods.

Market Externalities

- Public goods can also create externalities that affect markets. For example, investments in public infrastructure like roads and education can have positive spillover effects on private economic activity. Conversely, negative externalities such as pollution can arise when public goods are not properly managed, impacting market outcomes.

Market-based Solutions

- In some cases, market mechanisms can be used to address public goods provision. For instance, the creation of property rights or the use of market-based incentives like cap-and-trade systems can help internalize the externalities associated with public goods.

Types of Public Goods

Public goods are characterized by two key features: non-excludability and non-rivalrous consumption. Here are some types of public goods based on different criteria:

Pure Public Goods

- These are goods that are perfectly non-excludable and non-rivalrous in consumption. Once provided, individuals cannot be excluded from using them, and one person's consumption does not diminish the availability of the good for others. Examples include national defense, street lighting, and public parks.

Impure Public Goods

- These goods exhibit characteristics of public goods but may not fully meet the criteria of being non-excludable or non-rivalrous. For example, a congested public road may not be non-rivalrous during peak hours, and some people may be excluded from using certain public facilities due to capacity constraints.

Global Public Goods

- These are goods or services that are non-excludable and non-rivalrous in consumption on a global scale. They benefit everyone, regardless of national boundaries. Examples include efforts to combat climate change, control pandemics, and preserve biodiversity.

Local Public Goods

- These goods are non-excludable and non-rivalrous within a specific locality or community. Examples include local parks, public transportation systems, and community libraries.

Club Goods

- Also known as "excludable public goods" or "toll goods," club goods are excludable but non-rivalrous in consumption. They are typically provided by clubs or organizations, and individuals can be excluded from using them if they do not pay the required membership or subscription fee. However, once a person is a member, their consumption does not reduce the availability for others.
- Examples include cable television, private parks, and subscription-based online services.

Properties of Public Goods

- Public goods possess several key properties that distinguish them from other types of goods. These properties are essential for understanding the challenges associated with their provision and consumption.

Non-excludability vs Non-rivalrous Consumption

- Public goods are characterized by the inability to exclude individuals from consuming them once they are provided. It is difficult, if not impossible, to charge a price for the use of public goods and prevent non-paying individuals from benefiting from their consumption.

Non-rivalrous Consumption

- Consumption of a public good by one individual does not diminish its availability or utility for others. Unlike private goods, where consumption by one person reduces the quantity available for others, public goods can be enjoyed simultaneously by multiple individuals without affecting their enjoyment.

Jointness in Supply

- Public goods are typically jointly supplied, meaning that the cost of providing the good to one additional individual is zero or negligible. Once the infrastructure or system for providing the public good is in place, additional individuals can benefit from its provision without significantly increasing the cost.

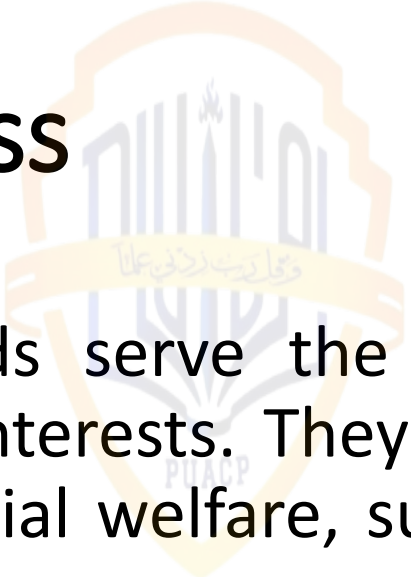
Difficulty in Pricing

- Due to their non-excludable and non-rivalrous nature, public goods present challenges in pricing. Since individuals cannot be excluded from using them and their consumption does not reduce availability for others, determining a fair price for access is complex.

Free Rider Problem

- The non-excludability of public goods leads to the free rider problem, where individuals have an incentive to consume the good without contributing to its provision. Since they can benefit from the good without paying, rational individuals may choose not to contribute, leading to under-provision or inadequate funding for the public good.

Publicness



- Public goods serve the collective interest of society rather than individual interests. They often have characteristics that make them vital for social welfare, such as national defense, clean air, or public infrastructure.